

2014-2015 – Master 2 – Macro I

Lecture 7 : Asset Values

Franck Portier

franck.portier@TSE-fr.eu

Toulouse School of Economics

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Changes from version 1.0 are in red

Changes from version 1.1 are in purple

Disclaimer

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1. Model

- ▶ Consider an asset which pays a dividend d_t
- ▶ Value at t *before dividends* is V_t .
- ▶ Assume that I can get a rate of return elsewhere equal to r .
- ▶ If I sell my asset now and invest the proceeds at rate r , I will get $(1 + r)V_t$ tomorrow.
- ▶ If I hold it, I will get $(1 + r)d_t + V_{t+1}$.
- ▶ Therefore the arbitrage condition is

$$V_t = d_t + \frac{V_{t+1}}{1 + r}. \quad (1)$$

- ▶ Equivalently we can write :

$$rV_t = (1 + r)d_t + (V_{t+1} - V_t). \quad (2)$$

1. Model

- That is :

Rate of Return x Value of the asset = Dividends +
Capital Gains.

- Note that for risk neutral agents, the formula extends to risky investments :

$$rV_t = (1 + r)d_t + E_t(V_{t+1} - V_t).$$

- That is :

Rate of Return x Value of the asset = Dividends +
Expected Capital Gains

1. Model

- ▶ Remark : if we consider value in period t *after dividends* $\tilde{V}_t$.
- ▶ If I sell my asset now and invest the proceeds at rate r , I will get $(1 + r)\tilde{V}_t$ tomorrow.
- ▶ If I hold it, I will get $d_{t+1} + \tilde{V}_{t+1}$ tomorrow.
- ▶ Therefore the arbitrage condition is

$$\tilde{V}_t = \frac{d_{t+1} + \tilde{V}_{t+1}}{1 + r}. \quad (3)$$

- ▶ Equivalently we can write :

$$r\tilde{V}_t = d_{t+1} + (\tilde{V}_{t+1} - \tilde{V}_t). \quad (4)$$

1. Model

- ▶ An equation like (1) can be solved by *forward integration*

$$V_t = \sum_{i=t}^T \frac{1}{(1+r)^{i-t}} d_i + \frac{V_{T+1}}{(1+r)^{T+1-t}}. \quad (5)$$

- ▶ Assume that the following sequence exists :

$$F_t = \sum_{i=t}^{+\infty} \frac{1}{(1+r)^{i-t}} d_i \quad (6)$$

- ▶ Then if $r > 0$ F_t is the only non explosive solution to (5).
- ▶ If we eliminate explosive solutions, then we have a unique one which is called the *fundamental* value of the asset.
- ▶ Furthermore, let $\tilde{V}_t$ be any other solution.
- ▶ Let $B_t = \tilde{V}_t - F_t$.
- ▶ It must be that

$$B_{t+1} = (1+r)B_t.$$

- ▶ Therefore, any solution is the sum of the "fundamental" and a "bubble" which explodes at rate r .

1. Model

- ▶ Remark : In fact the bubble can be stochastic and only need to explode at rate r in expectations. That is, up to a multiplicative factor $(1 + r)$ the bubble can be any martingale.
- ▶ Assume that the bubble can burst with probability p and that agents are risk neutral.
- ▶ Arbitrage requires $E_t B_{t+1} = (1 + r)B_t$
- ▶ Now (with an abuse of notation as I denote the stochastic variable and the realization by the same symbol B_{t+1})

$$E_t B_{t+1} = (1 - p)B_{t+1} + p \times 0$$

- ▶ Therefore $(1 - p)B_{t+1} + p \times 0 = (1 + r)B_t$, so that

$$\begin{aligned} B_{t+1} &= \frac{1 + r}{1 - p} B_t \\ &\approx (1 + r + p)B_t \quad \text{when it does not burst} \end{aligned}$$

1. Model

- Remark : Any present discounted value of an income (dividend) flow can be rewritten in a recursive fashion. For example if

$$V_t = \sum_{i=t}^{+\infty} \frac{1}{(1+r)^{i-t}} d_i, \quad (7)$$

- This can be written

$$V_t = d_t + \frac{1}{1+r} V_{t+1}. \quad (8)$$

Then this can be again interpreted in a dividend plus capital gains fashion :

$$rV_t = (1+r)d_t + V_{t+1} - V_t. \quad (9)$$

2. Continuous Time

Pricing the asset

- ▶ Suppose an asset yields a flow of dividends per unit of time equal to $x(t)$.
- ▶ Suppose I can get a return r per unit of time elsewhere.
- ▶ Let $V(t)$ be the value of the asset at t *before dividends*
- ▶ Suppose I sell it and invest in the market over a small time interval dt .
- ▶ Then at $t + dt$ I will have $V(t)(1 + r.dt)$.
- ▶ If instead I hold the asset, then at $t + dt$ I will have $x(t).dt(1 + r.dt) + V(t + dt)$.
- ▶ Hence *by arbitrage*

$$x(t).dt(1 + r.dt) + V(t + dt) = V(t)(1 + r.dt), \quad (10)$$

- ▶ Neglecting second order terms (*in dt^2*),
 $V(t + dt) \approx V(t) + \dot{V}(t)dt$
- ▶ so that (10) writes

$$rV(t) = x(t) + \dot{V}(t)$$

2. Continuous Time

Pricing the asset

- ▶ Compute now the value of the asset at t after dividends $\tilde{V}(t)$
- ▶ Suppose I sell it and invest in the market over a small time interval dt .
- ▶ Then at $t + dt$ I will have $\tilde{V}(t)(1 + r \cdot dt)$.
- ▶ If instead I hold the asset, then at $t + dt$ I will have $x(t + dt) \cdot dt + \tilde{V}(t + dt)$.
- ▶ Hence by arbitrage

$$\tilde{V}(t)(1 + r \cdot dt) = x(t + dt) \cdot dt + \tilde{V}(t + dt), \quad (11)$$

- ▶ Neglecting second order terms (in dt^2),
 $\tilde{V}(t + dt) \approx \tilde{V}(t) + \dot{\tilde{V}}(t)dt$ and $x(t + dt) \approx x(t) + \dot{x}(t)dt$
- ▶ so that (11) writes

$$\begin{aligned} \tilde{V}(t) + r\tilde{V}(t)dt &= (x(t) + \dot{x}(t)dt)dt + \tilde{V}(t) + \dot{\tilde{V}}(t)dt \\ r\tilde{V}(t)dt &= x(t)dt + \dot{\tilde{V}}(t)dt \end{aligned}$$

2. Continuous Time

Pricing the asset

- Note that in continuous time

$$\tilde{V}(t) = V(t)$$

2. Continuous Time

Pricing the asset

$$rV(t) = x(t) + \dot{V}(t)$$

- ▶ The LHS is the rate of return (per unit of time) times the value of the asset.
- ▶ The RHS is the dividend per unit of time, plus the expected capital gain per unit of time.
- ▶ The equation is the same as (2), except that the arbitrage takes place over a small time interval and that all is expressed per unit of time.
- ▶ In continuous time the fundamental is

$$F_t = \int_t^{+\infty} e^{-r(v-t)} x(v) dv$$

- ▶ And a bubble is such that

$$\dot{B} = rB \quad \text{that is} \quad B_t = B_0 e^{rt}.$$

2. Continuous time

Exogenous shocks

- ▶ Assume risk neutrality
- ▶ Assume that we have an asset whose value at date t is $V(t)$, pays dividends $x(t)$ and that this asset can exogenously be transformed into another asset.
- ▶ This other asset date t value is denoted $\hat{V}(t)$.
- ▶ Example : Consider that the asset is a car, whose value is V . With some probability, the owner has an accident and the car turns into a broken car, whose value is denoted $\hat{V}$
- ▶ The change of nature (the car accident) is Poisson distributed, with intensity $\lambda > 0$

2. Continuous time

Exogenous shocks

- ▶ Note : for a Poisson process with intensity $\lambda > 0$, the probability of having k events in time interval h is $\frac{e^{-\lambda h}(\lambda h)^k}{k!}$
- ▶ Note : when h goes to 0,
 - × Probability of observing 0 events in a time interval h
 $= e^{-\lambda h} \approx 1 - \lambda h$
 - × Probability of observing 1 events in a time interval h
 $= e^{-\lambda h} \lambda h \approx \lambda h$
 - × Probability of observing more than events in a time interval h
 ≈ 0

2. Continuous time

Exogenous shocks

- ▶ How to write the asset valuation equation ?
- ▶ Let $V(t)$ be the value of the asset at t .
- ▶ Suppose I sell it and invest in the market over a small time interval dt .
- ▶ Then at $t + dt$ I will have $V(t)(1 + r.dt)$.
- ▶ If instead I hold the asset, then at $t + dt$ I will have

$$x(t).dt + (1 - \lambda dt)V(t + dt) + \lambda dt \hat{V}(t + dt)$$

- ▶ Hence, by arbitrage

$$V(t)(1+r.dt) = x(t).dt(1 + r.dt) + (1 - \lambda dt)V(t + dt) + \lambda dt \hat{V}(t + dt)$$

2. Continuous time

Exogenous shocks

$$V(t)(1+r.dt) = x(t).dt(1 + r.dt) + (1-\lambda dt)V(t+dt) + \lambda dt \hat{V}(t+dt) \quad (\star)$$

- First order Taylor expansion :

$$\begin{cases} V(t+dt) = V(t) + \dot{V}(t)dt \\ \hat{V}(t+dt) = \hat{V}(t) + \dot{\hat{V}}(t)dt \end{cases}$$

- $(\star)$ becomes

$$\begin{aligned} V(t) + rV(t)dt &= x(t)dt + rx(t).dt^2 \\ &+ V(t) + \dot{V}(t)dt - \lambda V(t)dt - \dot{V}(t)(dt)^2 \\ &+ \lambda \hat{V}(t)dt + \dot{\hat{V}}(t)(dt)^2 \end{aligned}$$

- Simplify the gray terms, neglect the order two ones gives

$$rV(t) = x(t) + \lambda (\hat{V}(t) - V(t)) + \dot{V}(t)$$